

Features of DataHub--Smart Schedule



Background

In some countries (such as Germany, the United Kingdom, Belgium, and Australia), there are requirements to limit the feed-in power of PV systems. Any electricity exceeding the grid connection limit will be discarded.

Electricity prices generally exhibit fluctuations, with some countries experiencing significant variations. For example, the daily electricity prices in Finland can vary greatly, with the highest prices being over 6.8 times higher than the lowest prices, or even more.

Introduction

Smart Schedule is a system integrated in DataHub to flexibly adjust the inverter operating mode and system energy treatment based on the PV intensity, load forecast, and electricity price differences in a day, aiming to maximize the PV energy utility and revenue.

Features

1.AI-Driven Automation for Optimal Strategy Execution

AI-Driven Automation optimizes strategy execution with advanced algorithms, customizing energy treatment based on dynamic factors like PV intensity, electricity prices, and user habits. By analyzing these variables intelligently, it maximizes PV energy utility for efficiency and cost-effectiveness. This adaptive approach aligns with user preferences and environmental conditions, achieving optimal execution.

2.Home-Tailored Energy Solutions with Machine Learning

Home-Tailored Energy Solutions leverage Machine Learning for personalized energy optimization, providing unique solutions based on each home's usage habits. This advanced feature uses algorithms to analyze individual consumption patterns, adapting to household nuances. By learning from user behavior and environmental factors, it evolves to maximize efficiency and align with home-specific needs. This integration ensures dynamic and adaptive energy management, delivering a seamless and optimized experience for homeowners.

3.Boost Green Power 10% for a Greener Planet

When anticipating brighter sunlight from the previous day's forecast, there's an opportunity to start selling electricity earlier. By strategically using this early generation, surplus power typically curtailed during peak periods can recharge energy storage batteries. This approach optimizes energy use and enhances

overall green power output from photovoltaic sources.

4. Increase your economic efficiency by Max. 30%

Intelligent regulation, facilitated by AI, saves costs by avoiding grid power during nighttime peak electricity prices. This approach uses stored battery power from daylight hours to meet nighttime energy demand, reducing reliance on expensive grid power. By forecasting price fluctuations and dynamically managing consumption, the system optimizes resource allocation and enhances efficiency and sustainability.

How to Use

1. System Wiring Diagrams

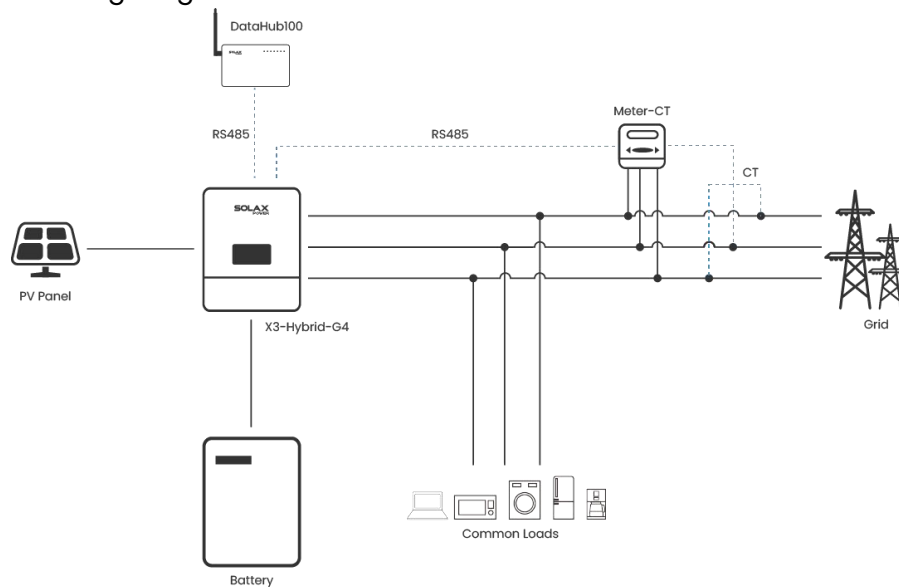


Figure 1 Standalone connection

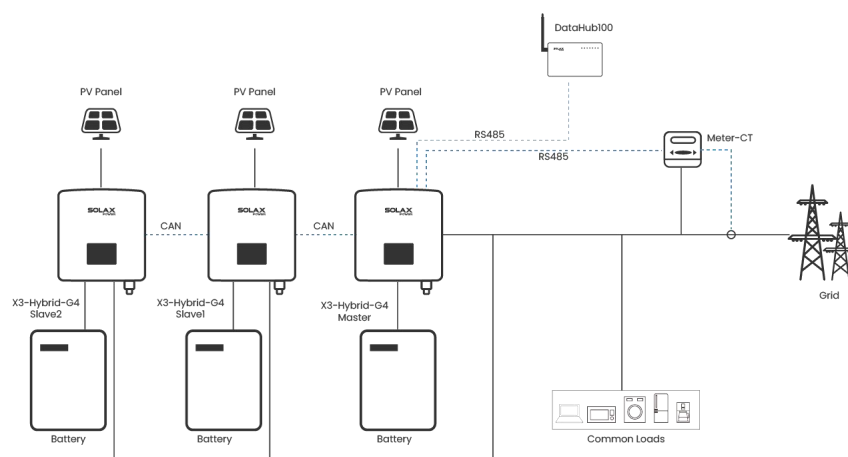


Figure 2 Parallel connection

2. Setting on the inverter

① Update the inverter software to the latest version. (Take X3 G4 Hybrid as an Example)

② Setting Path:

Settings > Advance Setting > Modbus, and then set Function Control to DataHub;

Settings > Advance Setting > Export Control.

Com485 address=1; Baud Rate=19200; User Value=defined limit value.

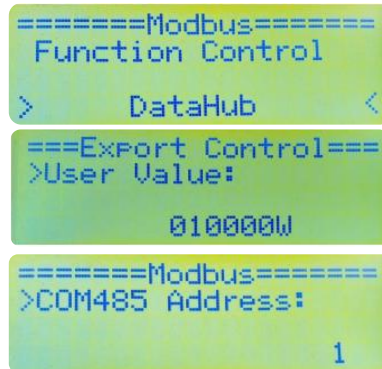


Figure 3 Setting on the inverter

③ Use a communication cable to connect the inverter COM terminal and DataHub RS485-1 terminal.

3. Creating Sites on SolaXCloud

① Go to www.solaxcloud.com and then enter the account user name and password to log in to SolaXCloud.

② On the homepage, click **System&Site > Add**. On the **Add Site Information** page, enter the site name, system size, country and region, and time zone.

③ Click  to open the map and select the site location, and then click Save.

④ Enter the SN or registration NO. of DataHub to bind to the site. You can click  to add more DataHub devices, and click  to delete it.

⑤ Select Use&Set DST if you need to. This option is not available in certain regions.

⑥ Select **I agree that site can be remotely maintained and functionally set by the agent/installer** and **I agree that this information will be visible to agents and installers**, and then click **Save**.

4. Setting on DataHub Webpage

① Connect your computer to DataHub hotspot named Wi-Fi_SN

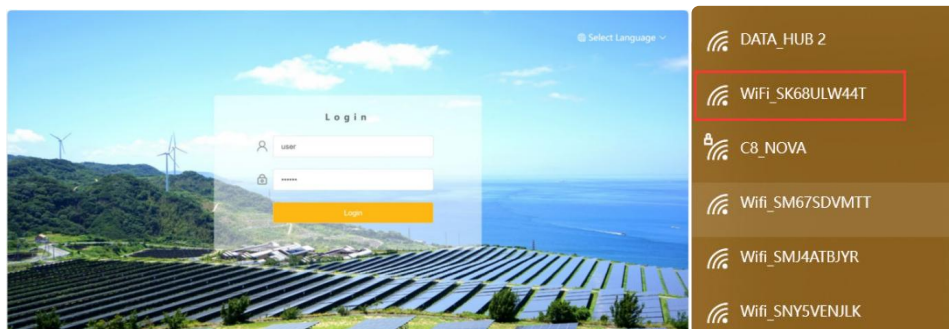


Figure 4 WLAN Page

- ② Go to web-page **192.168.10.10**, and then enter the account username and password to log in to the web-page

Username:: User

Password: 123456

- ③ On the homepage, select **DataHub Setting > Serial Port Setting** to check the baud rate. Make sure the baud rate of RS485 Channel 1 is 19200.

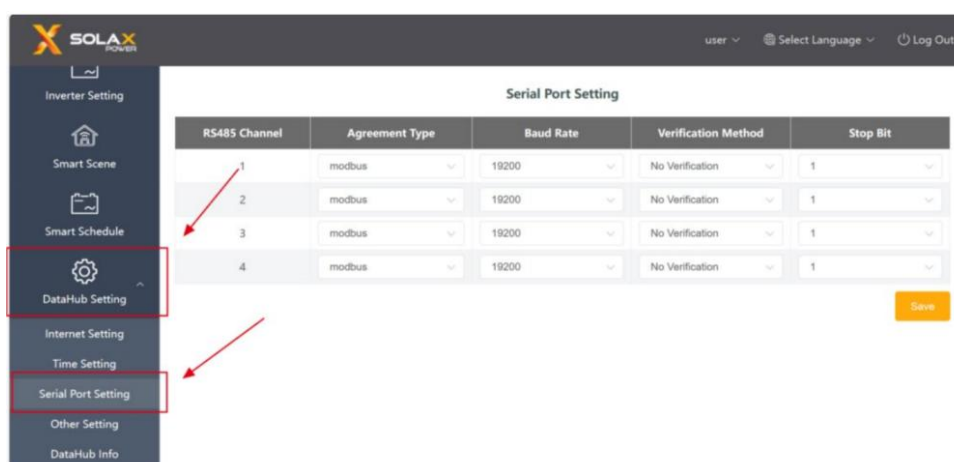


Figure 5 Setting Page

- ④ Select **Site Management > Add Device**, set **Device Type** of RS485 Channel 1 to **Inverter**. **Initial Address** to 1. **Number of Devices** to 1 or N based on the Connection situation and then click **Save**.

After the devices are added successfully, the device SN and model will be displayed on the prompt.

- ⑤ For LAN connection, connect the network cable to the NET port of DataHub. For Wi-Fi connection, select **DataHub Setting > Internet Setting**, and then enter the password to connect the Wi-Fi.

After the network is connected successfully, you can select **System Setting > System Info** to view the assigned IP address.

⑥ Click Smart Schedule click  to enable it, and then click Update to refresh the battery data.

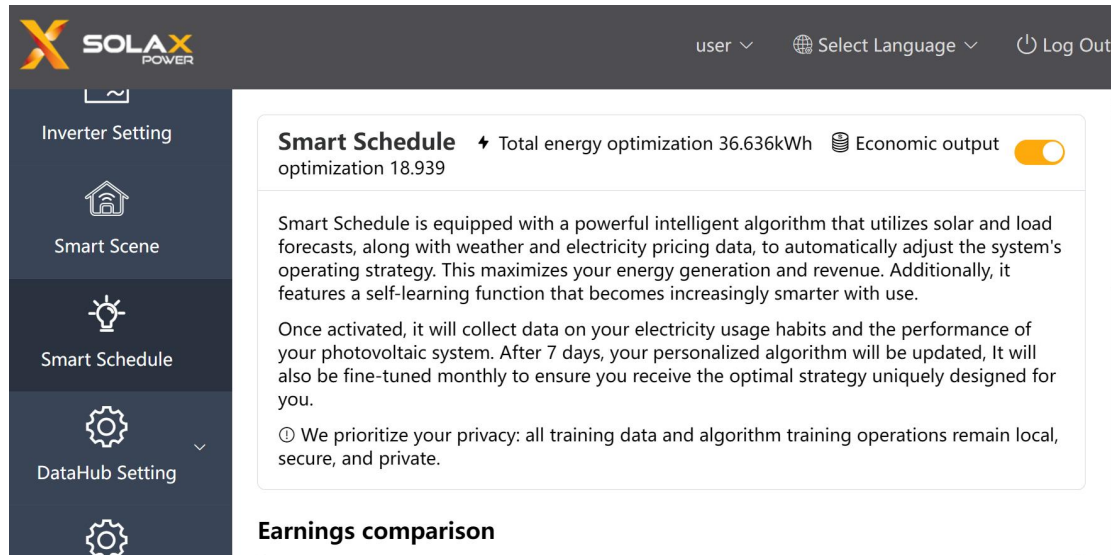


Figure 6 Enable Smart Schedule

⑦ Smart Schedule will capture electricity prices from the Cloud platform, adjusting strategies in real time to get the most out of them. On this page you can see the revenue comparison and the energy forecast

How it Works

1. Without Smart Schedule

In the constrained generation scenario, between 7:00 and 12:00, solar power from the PV system caters to the load requirements. Simultaneously, any **surplus energy is directed towards charging the battery**. Approaching noon, **the battery reaches its full capacity**. At 12:00, the excess solar power is efficiently fed back to the utility grid. However, due to the imposed restriction on power generation from the utility, any portion of the PV power exceeding the utility's capacity is discarded.

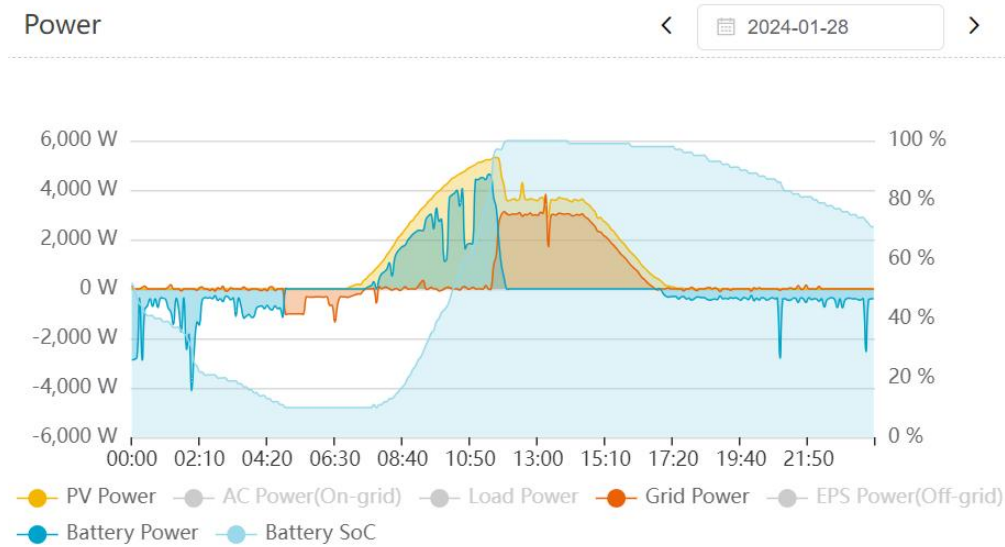


Figure 7 Power Diagram without Smart Schedule

2. With Smart Schedule

In the context of constrained generation between 7:00 and 13:00, the PV system initially caters to the load, with **surplus electricity sold to the grid**. Subsequently, the PV system prioritizes charging the battery, effectively utilizing electricity that would otherwise be wasted in a "without smart schedule" scenario, thereby augmenting overall power generation. Simultaneously, considering the typically higher electricity prices in the morning compared to midday, selling electricity during the morning hours enhances economic value.

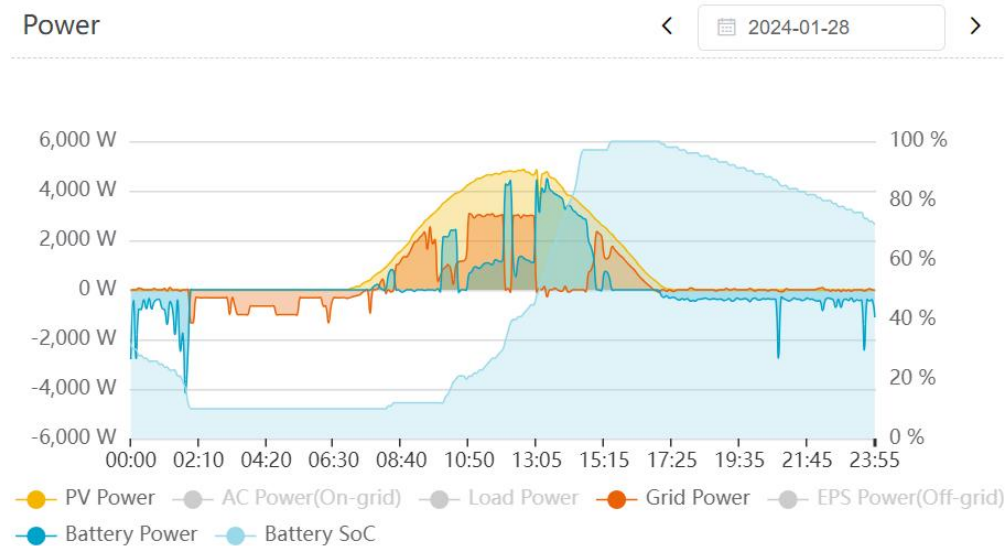


Figure 8 Power Diagram with Smart Schedule

Checking the Revenue

On the homepage, select Smart Schedule to view the feature results.

Table 1 Feature item description

Item	Description
Total Energy optimized	Excessive power that the system generates after Smart Schedule is enabled. The value is 0 under export control scenario.
Total Economy optimized	Excessive revenue that the system brings about combining the excessively generated energy and period-based electricity selling strategy.



Figure 9 Checking the Revenue

Capabilities

FUNCTION	INVERTER TYPE
Smart Schedule	X1-IES
	X3-IES
	X1-Hybrid G4
	X3-Hybrid G4
	X1-Fit G4



X1-Fit G4